KENYA MEDICAL TRAINING COLLEGE
FACULTY OF REHABILITATIVE SCIENCES
DEPARTMENT OF ORTHOPAEDIC TECHNOLOGY

FINAL QUALIFYING EXAMINATION
FOR
DIPLOMA IN ORTHOPAEDIC TECHNOLOGY

PAPER: PROSTHETICS AND ORTHOTICS

DATE: 6TH FEBRUARY 2025 **TIME: 3 HOURS (9:00AM – 12:00 NOON)**

INSTRUCTIONS

1. This paper consists of:
   - Section 1 (40 Multiple Choice Questions)
   - Section 2 (8 Short Answer Questions)
   - Section 3 (1 Long Answer Question)
2. Attempt **ALL** Questions
3. Write the EXAMINATION **NUMBER** given on all the answer sheets provided and on the question paper.
4. Ensure that all examination answer scripts are handed in at the end of the examination
5. Ensure you sign the examination register provided

EXAMINATION NUMBER .....

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SECTION ONE: MULTIPLE ANSWER QUESTIONS (40 MARKS)

1. Which level of amputation typically involves the removal of the foot at the ankle joint?
   A. Knee disarticulation
   B. Syme's amputation
   C. Transfemoral amputation
   D. Hip disarticulation
2. What are the biomechanical advantages of using a polycentric knee mechanism in transfemoral prostheses?
   A. Increased stability during stance phase
   B. Improved energy efficiency during swing phase
   C. Enhanced shock absorption during weight-bearing
   D. Reduced mechanical stress on the residual limb
3. In the context of prosthetic sockets, what is the purpose of incorporating relief areas or cutouts?
   A. To reduce overall weight of the prosthesis
   B. To enhance suspension of the prosthesis
   C. To accommodate bony prominences and sensitive areas of the residual limb
   D. To increase the durability of the socket
4. Which suspension system is most suitable for a highly active individual engaging in activities such as running and jumping?
   A. Silesian belt
   B. Vacuum-assisted suspension
   C. Pin lock mechanism
   D. Supracondylar cuff
5. How does dynamic alignment differ from static alignment in prosthetic limb fitting?
   A. Dynamic alignment involves adjusting the socket shape, while static alignment focuses on component alignment.
   B. Dynamic alignment considers the patient's gait pattern, while static alignment focuses on the anatomical landmarks.
   C. Dynamic alignment is performed during walking, whereas static alignment is conducted while the patient is stationary.
   D. Dynamic alignment aims to optimize energy expenditure, while static alignment prioritizes comfort and stability.
6. Which component of the lower limb prosthesis is responsible for absorbing shock and providing cushioning during weight-bearing activities?
   A. Pylon
   B. Foot
   C. Socket
   D. Suspension system
7. During gait training, what role does the prosthetist play in addressing pathological gait patterns?
   A. Prescribing medication to alleviate gait abnormalities
   B. Designing custom orthotic devices to correct gait deviations
   C. Analyzing gait biomechanics and adjusting prosthetic components as needed
   D. Administering physical therapy exercises to improve gait symmetry

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8. Which type of lower limb prosthesis is most appropriate for a patient with a high unilateral hip disarticulation?
   A. Syme's prosthesis
   B. Transfemoral prosthesis
   C. Transtibial prosthesis
   D. Knee disarticulation prosthesis
9. What is the goal of gait training for individuals with lower limb prostheses?
   A. Achieving maximum walking speed
   B. Minimizing energy expenditure during ambulation
   C. Restoring normal gait patterns and improving functional mobility
   D. Preventing falls and enhancing balance
10. Which pathological gait deviation is characterized by an exaggerated inward rotation of the foot during the swing phase?
   A. Equinus gait
   B. Trendelenburg gait
   C. Varus gait
   D. Drop foot gait
11. What distinguishes orthotics from prosthetics?
   A. Orthotics focus on replacing missing limbs, while prosthetics address musculoskeletal disorders.
   B. Orthotics are external devices used to support, align, prevent, or correct deformities, while prosthetics replace missing body parts.
   C. Orthotics primarily deal with lower limb disorders, while prosthetics focus on upper limb conditions.
   D. Orthotics are exclusively used for cosmetic purposes, while prosthetics are functional devices.
12. Which historical development significantly contributed to the advancement of orthotics?
   A. The discovery of anesthesia
   B. The invention of the X-ray machine
   C. The introduction of 3D printing technology
   D. The development of the first orthotic-specific materials
13. What biomechanical principle is utilized to correct a radial deviation deformity in upper limb orthotics?
   A. Plantarflexion
   B. Supination
   C. Pronation
   D. Dorsiflexion
14. What physiological consideration should you take into account when designing an orthotic device for a patient with carpal tunnel syndrome?
   A. The patient's cardiovascular fitness level
   B. The patient's age and gender
   C. The presence of nerve compression and tissue inflammation
   D. The patient's dietary habits

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15. Which orthotic component is commonly used to provide support and stability in the management of wrist drop?
    A. Plantarflexion stop
    B. Dorsiflexion assist
    C. Pronation block
    D. Supination assist
16. What aspect is assessed during the bench alignment process of an orthotic device?
    A. The patient's pain tolerance level
    B. The range of motion of the affected joint
    C. The alignment of the orthotic components in a static position
    D. The patient's emotional state and psychological well-being
17. What pathological gait deviation is commonly observed in patients with cerebral palsy and often requires orthotic intervention?
    A. Trendelenburg gait
    B. Antalgic gait
    C. Scissoring gait
    D. Equinus gait
18. Which type of upper limb orthosis is most suitable for correcting a flexion contracture of the elbow joint?
    A. Wrist cock-up splint
    B. Humeral fracture brace
    C. Elbow extension orthosis
    D. Thumb spica splint
19. What is the primary purpose of aligning orthotic components dynamically during gait analysis?
    A. To improve patient comfort
    B. To optimize energy expenditure
    C. To enhance orthotic durability
    D. To assess joint range of motion
20. What type of splint is commonly used to provide support and immobilization following a wrist fracture?
    A. Thumb spica splint
    B. Elbow extension splint
    C. Radial deviation splint
    D. Finger extension splint
21. What distinguishes upper limb prosthetics from lower limb prosthetics?
    A. Upper limb prosthetics focus on replacing missing limbs, while lower limb prosthetics address musculoskeletal disorders.
    B. Upper limb prosthetics are primarily cosmetic, while lower limb prosthetics are functional devices.
    C. Upper limb prosthetics involve more complex biomechanical considerations compared to lower limb prosthetics.
    D. Upper limb prosthetics are more commonly used in pediatric patients, while lower limb prosthetics are more prevalent in adults.

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22. Which historical development significantly contributed to the advancement of upper limb prosthetics?
    A. The invention of the X-ray machine.
    B. The introduction of 3D printing technology.
    C. The development of myoelectric control systems.
    D. The discovery of anesthesia.
23. What biomechanical principle is utilized to improve grasping and manipulation functions?
    A. Pronation and supination.
    B. Flexion and extension.
    C. Abduction and adduction.
    D. Rotation and translation.
24. Which level of upper limb amputation typically involves the removal of the entire hand and wrist joint?
    A. Transhumeral amputation.
    B. Transradial amputation.
    C. Wrist disarticulation.
    D. Forequarter amputation.
25. What is the primary purpose of the socket in upper limb prosthetics?
    A. To provide cosmetic restoration.
    B. To enhance proprioception.
    C. To transmit forces from the residual limb to the prosthesis.
    D. To improve psychological well-being.
26. Which suspension system is commonly used in upper limb prosthetics to secure the prosthesis to the residual limb?
    A. Vacuum-assisted suspension.
    B. Supracondylar cuff suspension.
    C. Pin lock mechanism.
    D. Silesian belt suspension.
27. Which component of the upper limb prosthesis is responsible for controlling elbow flexion and extension?
    A. Terminal device.
    B. Wrist unit.
    C. Elbow unit.
    D. Shoulder harness.
28. What is the primary consideration when selecting prosthetic components for pediatric upper limb amputees?
    A. Cosmetic appearance.
    B. Durability and longevity.
    C. Growth and development.
    D. Cost-effectiveness.
29. How does the shape and design of the prosthetic socket influence the comfort and function of the upper limb prosthesis?
    A. It minimizes energy expenditure during movement.
    B. It enhances proprioception and sensory feedback.
    C. It distributes pressure evenly across the residual limb.
    D. It facilitates cosmetic restoration and social integration.

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30. In cases of partial hand amputation, which type of upper limb prosthesis component is commonly used to restore prehensile functions?
    A. Passive cosmetic glove.
    B. Body-powered hook.
    C. Myoelectric terminal device.
    D. Cable-controlled wrist unit.
31. What distinguishes existing sports from specifically developed sports ni the context of adaptive sports for persons with disabilities?
    A. Existing sports are traditional sports that have been adapted for PWDs, while specifically developed sports are created exclusively for PWDs.
    B. Existing sports are those that PWDs can participate in without modifications, while specifically developed sports require adaptations to accommodate disabilities.
    C. Existing sports are played at the Paralympic Games, while specifically developed sports are not included in Paralympic competitions.
    D. Existing sports focus on individual performance, while specifically developed sports emphasize teamwork and cooperation among PWDs.
32. What is the primary goal of casting limbs in the context of adaptive sports prostheses?
    A. To create a cosmetic cover for the prosthetic limb.
    B. To improve the durability and longevity of the prosthetic limb.
    C. To optimize the fit and alignment of the prosthetic limb for sports performance.
    D. To enhance the aesthetics and appearance of the prosthetic limb during sports competitions.
33. How does the biomechanics of adaptive sports prostheses differ from traditional prostheses used in daily activities?
    A. Adaptive sports prostheses prioritize durability over performance.
    B. Adaptive sports prostheses are designed to enhance speed and agility.
    C. Adaptive sports prostheses focus on cosmetic appearance rather than functionality
    D. Adaptive sports prostheses are heavier and bulkier to provide stability during high-impact activities.
34. What medical considerations are important in the assessment of PWDs participating in wheelchair sports?
    A. Cardiovascular fitness and respiratory function.
    B. Psychological well-being and social integration.
    C. Dietary habits and nutritional status.
    D. Neurological deficits and sensory impairments.
35. How does the physiology of wheelchair sports athletes differ from able-bodied athletes?
    A. Wheelchair sports athletes have higher muscle endurance but lower cardiovascular capacity.
    B. Wheelchair sports athletes have similar physiological responses to able-bodied athletes.
    C. Wheelchair sports athletes may have impaired thermoregulation and heat dissipation.
    D. Wheelchair sports athletes are more prone to musculoskeletal injuries due to increased biomechanical stress.
36. Which type of wheelchair sport involves players using a racket to hit a shuttlecock over a net?
    A. Wheelchair basketball.
    B. Wheelchair rugby.
    C. Wheelchair tennis.
    D. Wheelchair racing.

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37. How does the set-up of a sport chair differ from a standard wheelchair?
    A. Sport chairs are heavier and bulkier to provide stability during high-speed maneuvers.
    B. Sport chairs have smaller wheels and a lower seat height for better aerodynamics.
    C. Sport chairs have adjustable seat angles and frame configurations for optimal performance.
    D. Sport chairs have fewer safety features to reduce weight and increase agility.
38. What role does exercise testing and prescription play in the training regimen of wheelchair sports athletes?
    A. It helps identify potential medical risks and contraindications for participation in sports.
    B. It determines the appropriate intensity and duration of training sessions for optimal performance.
    C. It evaluates the athlete's biomechanical efficiency and technique during sports activities.
    D. It assesses the psychological readiness and motivation of athletes for competitive sports.
39. How does footwear recommendations for adaptive sports differ from everyday footwear for PWDs?
    A. Adaptive sports footwear prioritizes cushioning and shock absorption over stability and support.
    B. Adaptive sports footwear is custom-designed to accommodate specific prosthetic components and alignment.
    C. Adaptive sports footwear has specialized traction patterns and outsole materials for various sports surfaces.
    D. Adaptive sports footwear focuses on fashion and aesthetics rather than performance and functionality.
40. What distinguishes the types of wheelchair sports based on the involvement of upper limb function?
    A. Wheelchair sports with upper limb involvement primarily use the lower body for propulsion.
    B. Wheelchair sports with upper limb involvement require athletes to have minimal upper limb function.
    C. Wheelchair sports with upper limb involvement emphasize throwing and catching activities.
    D. Wheelchair sports with upper limb involvement integrate both upper and lower body movements for performance.

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SECTION 2 SHORT ANSWER QUESTIONS (40 MARKS)

41. Explain the functional and biomechanical significance of five (5) parts of a shoe in the context of orthotic management and patient mobility. **(5 Marks)**
42. Explain the five (5) components of gait cycle in relation to transtibial prosthesis **(5 Marks)**
43. Compare five (5) types of prosthetic feet, focusing on their design features and functional outcomes for patient needs. **(5 Marks)**
44. Discuss types of prosthetic alignment and their impact on gait biomechanics and patient comfort. **(5 Marks)**
45. Discuss the role of orthotic devices in the rehabilitation of upper limb injuries or surgeries, considering the principles of biomechanics and patient-specific needs. **(5 Marks)**
46. Outline the important considerations when prescribing and fitting orthotic devices for pediatric patients with upper limb conditions, focusing on growth, development, and activity levels. **(5 Marks)**
47. Explain how the collaboration between orthotists, occupational therapists, and other healthcare professionals contributes to the holistic care of patients with upper limb conditions. **(5 Marks)**
48. Compare and contrast two (2) types of muscle contractions and three (3) implications for prosthetic and orthotic design. **(5 Marks)**

SECTION 3: LONG ANSWER QUESTION (20 MARKS)

49. Discuss management plan a for a diabetic patient with a worn out transtibial prosthesis focusing on the condition of the stump, condition of the prosthesis and recommended intervention **(20 Marks)**

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